

What's new in demand estimation?

Chris Conlon

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KU Leuven

What's new with micro data?

Data are a lot better than they were in 1995. It is common to observe:

- ▶ Choices of individuals, demographic characteristics, etc.
 - Examples: Nielsen Panelist Data, Kantar Worldpanel, IRI, etc.
- ▶ Higher frequency data, now $M_t \rightarrow \infty$ may not be a good assumption.
 - Not every product gets purchased every hour or every day.
 - Examples: Taxis/Uber, Daily airline bookings, hotel stays, AirBnB, Amazon, etc.

How can we make the best use of the data that we see?

1. Use control function approach of Petrin and Train (2010)
2. Fully specify likelihood with a parametric assumption on $f(\xi)$ (Jiang Machanda Rossi (2009))
3. Add “micro-moments” a la Petrin (2001) or Micro BLP (2004) but otherwise aggregate data.
 - How do we choose them? Are we throwing away information?
4. Add aggregate data moments to maximum likelihood (Grieco, Murry, Pinkse, Sagl 2023)

$$u_{ijt} = \beta_i x_{jt} - \alpha_i p_{jt} + \xi_{jt} + \varepsilon_{ijt}$$

$$p_{jt} = f(x_{jt}, z_{jt}) + e_{jt}$$

$$\xi_{jt} = \lambda e_{jt} + \zeta_{jt} \text{ with } \mathbb{E}[\zeta_{jt} \mid x_{jt}, z_{jt}] = 0$$

1. Run a first-stage for price to get $\hat{f}(\cdot)$ and $\hat{e}_{jt}$.
2. Plug in $\hat{e}_{jt}$ into inner IV regression of BLP:

$$\delta_{jt}(\theta_2) = x_{jt}\beta - \alpha p_{jt} + \lambda \hat{e}_{jt} + \zeta_{jt}$$

3. We can ignore λ , goal is to “clean” ξ_{jt} of the endogeneity and get $\hat{\alpha}$ correct.

Control Functions: What's the point?

- ▶ We can use the same $\hat{e}_{jt}$ approach and just do MLE like McFadden and Train instead. lambda, goal is to “clean” ξ_{jt} of the endogeneity.
- ▶ This means we can use full data on individual choices, while also addressing endogeneity of prices.
- ▶ Possibly more efficient but with really strong assumptions of $f(\cdot)$ being correctly specified
- ▶ Controversy: We need a consistent estimate for $\hat{e}_{jt}$ so $f(x_{jt}, z_{jt})$ needs to be the “true” mapping from observables to endogenous prices.

These approaches come from the Bayesian Marketing literature and “Bayesian IV”

1. Specify a reduced form pricing equation:

$$p_{jt} = \gamma z_{jt} + \omega_{jt}$$

2. Specify a parametric (joint) distribution for $f(\xi, \omega)$

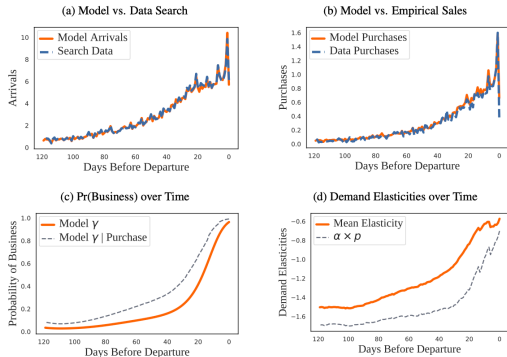
- Idea: be as flexible as possible allowing for correlation and mixtures of normals. (Bayesian Nonparametrics)
- Challenge: working out the Jacobian from $s_{jt} \rightarrow \xi_{jt}$ (just as in BLP).

3. Advantage: now we can do individual data MLE. Can also allow for very flexible distributions of $f(\beta_i | \theta)$ if we do MCMC.

Main Complaint: Markups are a function of **everything** including $\omega_{-j,t}$ and $\xi_{-j,t}$ (maybe?)

Fully specified $f(\xi, \omega)$: Hortacsu, Natan, Parsley, Schweig, Williams (2023)

Figure 6: Demand Estimates



Note: The horizontal axis of all plots denotes the negative time index, e.g. zero corresponds to the last day before departure. (a) Normalized model fit of searches with data searches. (b) Model fit of product shares with empirical shares. (c) Fitted values of γ_t over time, along with the probability a consumer is business conditional on purchase. (d) Mean product elasticities over time, along with the least and most elastic flights.

A nice extension/application of this approach

- Consider high-frequency airline data where markets M_t are **small**
- Allow for poisson arrivals of consumers to the market who choose with mixed logit probabilities
- Not much individual data. Allow for business/leisure travelers (unobserved types)
- Mixture of Dirichlet Process for (ξ, ω)
- One really good z_{jt} (estimated shadow prices)

Combined Likelihood Approaches (GMPS 2023)

- The researcher observes market-level data of purchases, and can construct market shares

$$s_{jm} = \frac{1}{N_m} \sum_{i=1}^{N_m} d_{ijm} \quad (1)$$

where N_m is the market size.

- For a subset of S_m consumers (denoted by the dummy D_{im}) the researcher observes $\{(d_{i \cdot m}, y_{im})\}$

Previous Approach: two-step estimator

- Estimate $(\hat{\delta}, \hat{\theta})$ by maximizing the likelihood

$$\mathcal{L}(\delta, \theta) = \sum_i \sum_j \sum_m d_{ijm} \log \int \frac{\exp(\delta_{jm} + \mu_{ijm}^y + \mu_{ijm}^\nu)}{\underbrace{\sum_{l=0}^{J_m} \exp(\delta_{jm} + \mu_{ijm}^y + \mu_{ijm}^\nu)}_{\sigma_{jm}(y_{im}, \nu; \theta, \delta)}} dF(\nu)$$

and recover $\hat{\beta}$ by running IV regression on $\hat{\delta}$.

Bayer, Ferreira, and McMillan (2007) Bayer and Timmins (2007)

- Uses only within-market variation to estimate θ , which can break identification.

- GMPS 2023 propose the following estimator:

$$(\hat{\beta}, \hat{\theta}, \hat{\delta}) = \arg \min_{\beta, \theta, \delta} \underbrace{(-\log \hat{L}(\theta, \delta) + \hat{\Pi}(\beta, \delta))}_{\hat{\Omega}(\beta, \theta, \delta)} \quad (2)$$

where

1. the **MDLE**

$$\log \hat{L}(\theta, \delta) = \sum_{m=1}^M \sum_{j=0}^{J_m} \sum_{i=1}^{N_m} d_{ijm} (D_{im} \log \sigma_{jm}^{y_{im}}(\theta, \delta) + (1 - D_{im}) \log \sigma_{jm}(\theta, \delta)) \quad (3)$$

$$= \sum_{m=1}^M \sum_{j=0}^{J_m} \sum_{i=1}^{N_m} D_{im} d_{ijm} \log \frac{\sigma_{jm}^{y_{im}}}{\sigma_{jm}} + \sum_{m=1}^M N_m \sum_{j=0}^{J_m} s_{jm} \log \sigma_{jm} \quad (4)$$

If $N_m = S_m$ this estimator $\leftrightarrow$ mixed-logit estimator in previous slides.

If $S_m = 0$ maximizing the likelihood $\leftrightarrow$ imposing share constraints

2. the **product level moments**

$$\hat{\Pi}(\beta, \delta) = \frac{1}{2} \hat{m}^\top(\beta, \delta) \hat{\mathcal{W}} \hat{m}^\top(\beta, \delta) \quad (5)$$

where $\hat{m}^\top(\beta, \delta) = \sum_{m=1}^M \sum_{j=1}^{J_m} b_{jm}(\delta_{jm} - \beta^\top x_{jm})$.

If $d_b = d_\beta$ (exact identification) $\leftrightarrow$ two-step estimator

If $d_b > d_\beta$ (overidentification): both terms contribute to the estimation of θ, δ

- Show asymptotic equivalence of MDPLE with the GMM estimator

$$(\hat{\beta}, \hat{\theta}, \hat{\delta}) = \arg \min_{\beta, \theta, \delta} \frac{1}{2} \begin{bmatrix} \hat{m}^\top & \partial_{\psi^\top} \log \hat{L} \end{bmatrix} \begin{bmatrix} \hat{\mathcal{W}} & 0 \\ 0 & \hat{\mathcal{W}}_L \end{bmatrix} \begin{bmatrix} \hat{m} \\ \partial_{\psi} \log \hat{L} \end{bmatrix} \quad (6)$$

- Show that the estimator above is efficient
- Trick: Also setup problem to be convex in δ

Properties: conformant convergence

- Conformant: convergence rates adjust depending on alternative divergence rates of $\{N_m\}$, S , J and variation in the data
- β is identified only from $\hat{\Pi}$: convergence rate is always $\sqrt{J}$.
- Converge rates for θ , δ depend on whether θ_z is fixed, and whether it is 0 or $\neq 0$.

case	rate		contributing term(s)	
	θ^z	θ^ν, δ	for θ^z	for θ^ν
$S/J \rightarrow \infty, \theta^z \neq 0$	$\sqrt{S}$	$\sqrt{S}$	$\log \hat{L}$	$\log \hat{L}$
$S/J \rightarrow \infty, \theta^z = 0$	$\sqrt{S}$	$\sqrt{J}$	$\log \hat{L}$	$\hat{\Pi}$
$S/J \rightarrow c, \theta^z \neq 0$	$\sqrt{J}$	$\sqrt{J}$	both	both
$S/J \rightarrow c, \theta^z = 0$	$\sqrt{J}$	$\sqrt{J}$	both	$\hat{\Pi}$
$S/J \rightarrow 0$	$\sqrt{J}$	$\sqrt{J}$	$\hat{\Pi}$	$\hat{\Pi}$

- Variation in the micro data alone is sufficient to identify θ_z , θ^ν and δ if the micro sample is large enough and demographic variation affects choice probabilities substantially.

Limitations of the MDPLE estimator

- ▶ Incorporating supply-side moment conditions, or other moments that may make the MDPLE objective function non-convex.
- ▶ Limited gains compared to other approaches (in particular, the Optimal Micro BLP Estimator) when $\frac{S_m}{N_m} \rightarrow 0$ as $N \rightarrow \infty$.
- ▶ Assumes fully-compatible dataset (not true, e.g., if the researcher has censored data).

Micro BLP Approaches

Micro BLP is used a lot

Paper	Industry	Paper	Industry
Petrin2002)	Automobiles	Barwick, Cao, and Li2017)	Automobiles
Berry, Levinsohn, and Pakes2004)	Automobiles	Murry2017)	Automobiles
Thomadsen2005)	Fast Food	Wollmann2018)	Commercial Vehicles
Goeree2008)	Personal Computers	S. Li2018)	Automobiles
Ciliberto and Kuminoff2010)	Cigarettes	Y. Li, Gordon, and Netzer2018)	Digital Cameras
Nakamura and Zerom2010)	Coffee	Backus, Conlon, and Sinkinson2021)	Cereal
Beresteanu and Li2011)	Automobiles	Grieco, Murry, and Yurukoglu2021)	Automobiles
S. Li2012)	Automobiles	Neilson2021)	Primary Schools
Copeland2014)	Automobiles	Armitage and Pinter2022)	Automobiles
Starc2014)	Health Insurance	Döpper, MacKay, Miller, and Stiebale2022)	Retail
Ching, Hayashi, and Wang2015)	Nursing Homes	Bodéré2023)	Preschools
S. Li, Xiao, and Liu2015)	Automobiles	Montag2023)	Laundry Machines
Nurski and Verboven2016)	Automobiles	Conlon and Rao2023)	Distilled Spirits
⋮	⋮	⋮	⋮

► Many empirical IO papers use the “micro BLP” approach

1. Impose the **Berry, Levinsohn, and Pakes1995)** share constraint (unlike Grieco et al. 2023)
2. Stack product-level or “**aggregated**” moments with “**micro**” moments from consumer surveys

Towards a standardized framework

- ▶ Despite the popularity of micro BLP, there's **no standardized framework**
 - Most papers use different notation to incorporate micro data in problem-specific ways
- ▶ So econometric work is limited, with a few exceptions
 - **Berry and Haile2014, 2022**): Nonparametric identification, including with micro data
 - **Myojo and Kanazawa2012**): Extend **Berry, Linton, and Pakes2004**) asymptotics to **Petrin2002**)
- ▶ We extend our BLP “best practices” (**Conlon & Gortmaker2020**) to the case with micro data
 1. Provide a **standardized framework** that covers most cases
 2. Derive **practical advice** from econometrics, simulations, examples
 3. Make all this easy to do with **PyBLP**

A standardized framework (Conlon Gortmaker 2023)

- ▶ Aggregate data generated market-by-market t
 - **Products** $j \in \mathcal{J}_t$ have observed characteristics x_{jt} , unobserved quality ξ_{jt}
 - **Consumer types** $i \in \mathcal{I}_t$ have observed demographics y_{it} , unobserved preferences ν_{it}
 - **Market shares** $\sigma_{jt} = \sum_i w_{it} \sigma_{ijt}$ integrate over consumer mass, each type has known weight w_{it}
- ▶ Micro data generated dataset-by-dataset d , conditional on aggregate data
 - Results $\{(t_n, j_n, y_{i_n t_n})\}_{n \in \mathcal{N}_d}$ from **independent surveys** of **selected consumers**
 - Each consumer n was surveyed with known probability $w_{di_n j_n t_n}$
- ▶ Often only have or willing to use **summary stats** (cost, compatibility, interpretability, etc.)
 - Smooth functions $f(\bar{v}_d)$ of averages $\bar{v}_d = \frac{1}{N_d} \sum_n v_{di_n j_n t_n}$
- ▶ “I want to match the mean demographic of consumers who purchased a product”
 - “ $\mathbb{E}[y_{it} \mid j \neq 0]$ ” $\leftarrow$ Let $w_{dijt} = 1\{j \neq 0\}$ and $v_{dijt} = y_{it}$

$$u_{ijt} = x'_{jt}(\beta_0 + \Pi_0 y_{it} + \Sigma_0 \nu_{it}) + \xi_{jt} + \varepsilon_{ijt}$$

- ▶ With only product-level aggregate data, often difficult to accurately estimate Π_0 and Σ_0
 - Often limited **cross-market variation** in demographic distributions and choice sets
- ▶ What **within-market** micro variation is informative about Π_0 ?
 - Literature tends to match stats that look like " $\mathbb{C}(x_{jt}, y_{it} \mid j \neq 0)$ "
- ▶ What about Σ_0 ?
 - Literature emphasizes **second choices**, e.g. " $\mathbb{C}(x_{jt}, x_{k(-j)t} \mid j, k \neq 0)$ "
- ▶ What about β_0 ?
 - Only **indirectly**: β_0 enters σ_{ijt} only through $\delta_{jt} = x'_{jt}\beta_0 + \xi_{jt}$, pinned down by share constraint

Support for most cases

Paper Micro moments shorthand

Petrin2002	$\mathbb{P}(j \in \mathcal{J} \mid i \in \mathcal{J}), \mathbb{E}[y_i \mid j \in \mathcal{J}]$
Berry et al. (2004)	$\mathbb{C}(x_j, y_i \mid j \neq 0), \mathbb{C}(x_j, x_{k(-j)} \mid j, k \neq 0)$
Thomadsen2005	$\mathbb{P}(j \in \mathcal{J} \mid i \in \mathcal{J})$
Goeree2008	$\mathbb{P}(j \in \mathcal{J} \mid i \in \mathcal{J})$
Ciliberto and Kuminoff2010	$\mathbb{E}[y_i \mid j \in \mathcal{J}]$
Nakamura and Zerom2010	$\mathbb{P}(j \in \mathcal{J} \mid i \in \mathcal{J})$
Beresteanu and Li2011	$\mathbb{P}(j \in \mathcal{J} \mid i \in \mathcal{J})$
S. Li2012	$\mathbb{P}(j \in \mathcal{J} \mid i \in \mathcal{J}), \mathbb{E}[y_i \mid j \in \mathcal{J}]$
Copeland2014	$\mathbb{E}[y_i \mid j \in \mathcal{J}]$
Starc2014	$\mathbb{P}(j \in \mathcal{J} \mid i \in \mathcal{J}), \mathbb{E}[x_j \mid i \in \mathcal{J}, j \neq 0]$
Ching et al.2015	$\mathbb{P}(j \in \mathcal{J} \mid i \in \mathcal{J})$
S. Li et al.2015	$\mathbb{P}(j \in \mathcal{J} \mid i \in \mathcal{J})$
Nurski and Verboven2016	$\mathbb{E}[y_i \mid j \in \mathcal{J}], \mathbb{C}(x_j, y_i \mid j \neq 0)$
	$\vdots \vdots$

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Y. Li et al.2018	$\mathbb{P}(j \in \mathcal{J} \mid i \in \mathcal{J})$
Backus et al.2021	$\mathbb{E}[y_i \mid j \in \mathcal{J}], \mathbb{C}(x_j, y_i \mid j \neq 0)$
Grieco et al.2021	$\mathbb{E}[x_j \mid i \in \mathcal{J}, j \neq 0], \mathbb{C}(x_j, x_{k(-j)} \mid j, k \neq 0)$
Neilson2021	$\mathbb{E}[x_j \mid i \in \mathcal{J}, j \neq 0]$
Armitage and Pinter2022	$\mathbb{E}[y_i \mid j \in \mathcal{J}]$
Döpfer et al.2022	$\mathbb{E}[y_i \mid j \in \mathcal{J}]$
Bodéré2023	$\mathbb{P}(j \in \mathcal{J} \mid i \in \mathcal{J}), \mathbb{E}[x_j \mid i \in \mathcal{J}, j \neq 0]$
Montag2023	$\mathbb{C}(x_j, y_i \mid j \neq 0), \mathbb{C}(x_j, x_{k(-j)} \mid j, k \neq 0)$
Conlon and Rao2023	$\mathbb{E}[y_i \mid j \in \mathcal{J}], \mathbb{E}[x_j \mid i \in \mathcal{J}, j \neq 0]$
	$\vdots \vdots$

► Framework supports most cases we've seen

- Demographic/choice-based sampling, conditioning, covariances, **second choices** $k \neq j$ too!

Optimal micro moments

- What in the full micro data $\{(t_n, j_n, y_{i_n t_n})\}_{n \in \mathcal{N}_d}$ is **most informative** about θ_0 ? The **score**!

$$v_{ijt}(\theta_0) = \frac{\partial \log \mathbb{P}(t_n = t, j_n = j, y_{i_n t_n} = y_{it} \mid n \in \mathcal{N}_d)}{\partial \theta'_0}$$

- Inspecting score expressions **gives intuition** for which micro moments perform well

$$\underbrace{v_{ijt}(\Pi_0) \approx x_{jt}y_{it}}_{\text{Similar to "}\mathbb{C}(x_{jt}, y_{it} \mid j \neq 0)\text{"}} \quad \underbrace{v_{ijkt}(\Sigma_0) \approx x_{jt} + x_{k(-j)t}}_{\text{Similar to "}\mathbb{C}(x_{jt}, x_{k(-j)t} \mid j, k \neq 0)\text{"}} \quad \underbrace{v_{ijt}(\beta_0) = 0}_{\text{No direct info}}$$

- Feasible to match $v_{ijt}(\hat{\theta})$ at consistent $\hat{\theta}$ in second GMM step (with optimal weights/IVs)
- **Asymptotically efficient** among all **share-constrained** micro BLP estimators
 - **Computationally efficient** too, only need to compute scores once
 - Must observe and be willing to use all info in the full micro data!

Some words of caution

- ▶ There are no efficiency guarantees for **inconsistent** pilot estimates $\hat{\theta}$
 - For first step, can use standard moments or score at informed guess of θ_0
- ▶ Most pairs of datasets have at least some **incompatibilities** in timing, variables, etc.
 - Optimal micro moments will only work well if incompatibilities are small
 - If large, match moments you expect to be compatible, e.g. correlations if scales are different
- ▶ Quadrature behaves poorly with **discontinuities** in moments like “ $\mathbb{E}[x_{jt} \mid y_{it} < \bar{y}, j \neq 0]$ ”
 - Instead, use Monte Carlo methods or moments continuous in y_{it} like “ $\mathbb{C}(x_{jt}, y_{it} \mid j \neq 0)$ ”

$$\hat{\theta} = \arg \min_{\theta} \hat{g}(\theta)' \hat{W} \hat{g}(\theta), \quad \hat{g}(\theta) = \begin{bmatrix} \hat{g}_A(\theta) \\ \hat{g}_M(\theta) \end{bmatrix} = \begin{bmatrix} \frac{1}{N_A} \sum_t \sum_j (\hat{\delta}_{jt}(\Pi, \Sigma) - x'_{jt} \beta) z_{jt} \\ f_1(\bar{v}) - f_1(v(\Pi, \Sigma)) \\ \vdots \\ f_{M_M}(\bar{v}) - f_{M_M}(v(\Pi, \Sigma)) \end{bmatrix}$$

- **Berry, Levinsohn, and Pakes's 1995** share constraint gives mean utilities $\hat{\delta}_{jt}(\Pi, \Sigma)$

$$s_{jt} = \sum_i w_{it} \frac{\exp[\hat{\delta}_{jt}(\Pi, \Sigma) + x'_{jt}(\Pi y_{it} + \Sigma \nu_{it})]}{1 + \sum_k \exp[\hat{\delta}_{kt}(\Pi, \Sigma) + x'_{kt}(\Pi y_{it} + \Sigma \nu_{it})]}$$

- Micro moments m match smooth functions $f_m(\cdot)$ of simple averages, called micro parts p

$$\bar{v}_p = \frac{1}{N_{d_p}} \sum_n v_{p i_n j_n t_n} \xrightarrow{P_A} v_p(\theta_0) = \mathbb{E}_A[v_{p i_n j_n t_n}] = \frac{\sum_t \sum_i \sum_j w_{it} \sigma_{ijt}(\theta_0) w_{d_p ijt} v_{pijt}}{\sum_t \sum_i \sum_j w_{it} \sigma_{ijt}(\theta_0) w_{d_p ijt}}$$

Collecting second choices for an empirical example

- ▶ We demonstrate how all this works with **Nielsen data**
 - Estimate pre-2017 demand for soft drinks in Seattle
- ▶ Counterfactual highlights **diversion to the outside good**
 - Predict effects of 2018 tax, compare with what happened (—22%)
- ▶ So also show how to run cheap **second choice survey**
 - Diversion ratios discipline counterfactual in interpretable way

	Scanner Data	& Household	& Diversion
Taxed Volume Change (%)	—30.1 (1.4)	—30.0 (1.5)	—16.5 (1.7)
▶ High — Low Income (pp)		2.0 (0.8)	1.0 (0.9)

If non-diet Coke was *not* available, what type of drink would you have purchased instead?

☐ Non-diet Pepsi  pepsi	☐ Diet Pepsi  diet pepsi
☐ Non-diet Gatorade  GATORADE	☐ Gatorade Zero 
☐ No drink or non-soft drink	☐ Diet Coke 
☐ Other non-diet soft drink	☐ Other diet soft drink

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Replicating Petrin (2002) with PyBLP

```
import numpy as np
import pandas as pd
from pyblp import data, Problem, Formulation, MicroDataset, MicroPart, MicroMoment, Optimization, Iteration

# Configure the aggregate problem: linear demand ("X1"), nonlinear demand ("X2"), marginal costs ("X3"), and demographics
problem = Problem(
    product_formulations=[
        Formulation('1 + hpwt + space + air + mpd + fud + mi + sw + su + pv + pgnp + trend + trend2'),
        Formulation('1 + I(-prices) + hput + space + air + mpd + fud + mi + sw + su + pv'),
        Formulation('1 + log(hput) + log(wt) + log(mpg) + air + fud + trend + (jp + eu) + log(q)'),
    ],
    costa_type='log',
    agent_formulation=Formulation('1 + I(low / income) + I(mid / income) + I(high / income) + I(log(fs) + fv) + age + fs + mid + high'),
    product_data=pd.read_csv(data.PETRIN_PRODUCTS_LOCATION),
    agent_data=pd.read_csv(data.PETRIN_AGENTS_LOCATION),
)

# Configure the micro dataset: name, number of observations, and a function that computes sampling weights
dataset = MicroDataset("CEX", 29125, lambda t, p, a: np.ones((a.size, 1 + p.size)))

# Configure micro moment parts: names, datasets, and functions that compute micro values
age_mi_part = MicroPart("E[age_i * mi_j]", dataset, lambda t, p, a: np.outer(a.demographics[:, 5], np.r_[0, p.X2[:, 7]]))
age_sw_part = MicroPart("E[age_i * sw_j]", dataset, lambda t, p, a: np.outer(a.demographics[:, 5], np.r_[0, p.X2[:, 8]]))
age_su_part = MicroPart("E[age_i * su_j]", dataset, lambda t, p, a: np.outer(a.demographics[:, 5], np.r_[0, p.X2[:, 9]]))
age_pv_part = MicroPart("E[age_i * pv_j]", dataset, lambda t, p, a: np.outer(a.demographics[:, 5], np.r_[0, p.X2[:, 10]]))
fs_mi_part = MicroPart("E[fs_i * mi_j]", dataset, lambda t, p, a: np.outer(a.demographics[:, 6], np.r_[0, p.X2[:, 7]]))
fs_sw_part = MicroPart("E[fs_i * sw_j]", dataset, lambda t, p, a: np.outer(a.demographics[:, 6], np.r_[0, p.X2[:, 8]]))
fs_su_part = MicroPart("E[fs_i * su_j]", dataset, lambda t, p, a: np.outer(a.demographics[:, 6], np.r_[0, p.X2[:, 9]]))
fs_pv_part = MicroPart("E[fs_i * pv_j]", dataset, lambda t, p, a: np.outer(a.demographics[:, 6], np.r_[0, p.X2[:, 10]]))
inside_mid_part = MicroPart("E[i{j > 0} * mid_i]", dataset, lambda t, p, a: np.outer(a.demographics[:, 7], np.r_[0, p.X2[:, 0]]))
inside_high_part = MicroPart("E[i{j > 0} * high_i]", dataset, lambda t, p, a: np.outer(a.demographics[:, 7], np.r_[0, p.X2[:, 1]]))
mi_part = MicroPart("E[mi_j]", dataset, lambda t, p, a: np.outer(a.demographics[:, 8], np.r_[0, p.X2[:, 7]]))
sw_part = MicroPart("E[sw_j]", dataset, lambda t, p, a: np.outer(a.demographics[:, 8], np.r_[0, p.X2[:, 8]]))
su_part = MicroPart("E[su_j]", dataset, lambda t, p, a: np.outer(a.demographics[:, 8], np.r_[0, p.X2[:, 9]]))
pv_part = MicroPart("E[pv_j]", dataset, lambda t, p, a: np.outer(a.demographics[:, 8], np.r_[0, p.X2[:, 10]]))
mid_part = MicroPart("E[mid_i]", dataset, lambda t, p, a: np.outer(a.demographics[:, 7], np.r_[1, p.X2[:, 0]]))
high_part = MicroPart("E[high_i]", dataset, lambda t, p, a: np.outer(a.demographics[:, 7], np.r_[1, p.X2[:, 1]]))

# Configure micro moments: names, observed values, parts, and functions that combine parts
ratio = lambda v: v[0] / v[1]
gradient = lambda v: [1 / v[1], -v[0] / v[1]**2]
micro_moments = [
    MicroMoment("E[age_i | mi_j]", 0.783, [age_mi_part, mi_part], ratio, gradient),
    MicroMoment("E[age_i | sw_j]", 0.730, [age_sw_part, sw_part], ratio, gradient),
    MicroMoment("E[age_i | su_j]", 0.740, [age_su_part, su_part], ratio, gradient),
    MicroMoment("E[age_i | pv_j]", 0.652, [age_pv_part, pv_part], ratio, gradient),
    MicroMoment("E[fs_i | mi_j]", 3.86, [fs_mi_part, mi_part], ratio, gradient),
    MicroMoment("E[fs_i | sw_j]", 3.17, [fs_sw_part, sw_part], ratio, gradient),
    MicroMoment("E[fs_i | su_j]", 2.97, [fs_su_part, su_part], ratio, gradient),
    MicroMoment("E[fs_i | pv_j]", 3.47, [fs_pv_part, pv_part], ratio, gradient),
    MicroMoment("E[i{j > 0} | mid_i]", 0.0794, [inside_mid_part, mid_part], ratio, gradient),
    MicroMoment("E[i{j > 0} | high_i]", 0.1581, [inside_high_part, high_part], ratio, gradient),
]

# Configure two-step minimum distance: starting values, optimization, and micro moments
problem_results = problem.solve(
    sigma=np.diag([3.23, 0, 4.43, 0.46, 0.01, 2.58, 4.42, 0, 0, 0, 0]),
    pi=np.array([
        [0, 0, 0, 0, 0, 0, 0, 0, 0, 0],
        [0, 7.52, 31.13, 34.49, 0, 0, 0, 0, 0],
        [0, 0, 0, 0, 0, 0, 0, 0, 0],
        [0, 0, 0, 0, 0, 0, 0, 0, 0],
        [0, 0, 0, 0, 0, 0, 0, 0, 0],
        [0, 0, 0, 0, 0, 0, 0, 0, 0],
        [0, 0, 0, 0, 0.57, 0, 0, 0, 0],
        [0, 0, 0, 0, 0.28, 0, 0, 0, 0],
        [0, 0, 0, 0, 0.31, 0, 0, 0, 0],
        [0, 0, 0, 0, 0.42, 0, 0, 0, 0],
    ]),
    optimization=Optimization('bfgs', {'gtol': 1e-4}),
    iteration=Iteration('squarem', {'atol': 1e-13}),
    se_type='clustered',
    W_type='clustered',
    micro_moments=micro_moments,
)
```

► We've tried our best to make all this easy to do with **PyBLP**, including optimal micro moments

- E.g. can estimate **Petrin's(2002)** model with < 100 lines of code

- ▶ If you have relatively complete data on individual decisions, do MDLE
- ▶ If you have mostly aggregate data and some survey moments do microBLP.
- ▶ If you have very high-frequency data with lots of zeros, you are probably stuck doing some full likelihood Bayesian approach.

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